For Examiner's Use

1 A student is investigating how the resistance R of nichrome in the form of a wire varies with temperature θ.

It is suggested that

$$R = R_0(1 + \alpha\theta)$$

where R_0 is the resistance at 0 °C, α is a constant and θ is the temperature measured in °C.

Design a laboratory experiment to test the relationship between θ and R and determine the value of α. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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